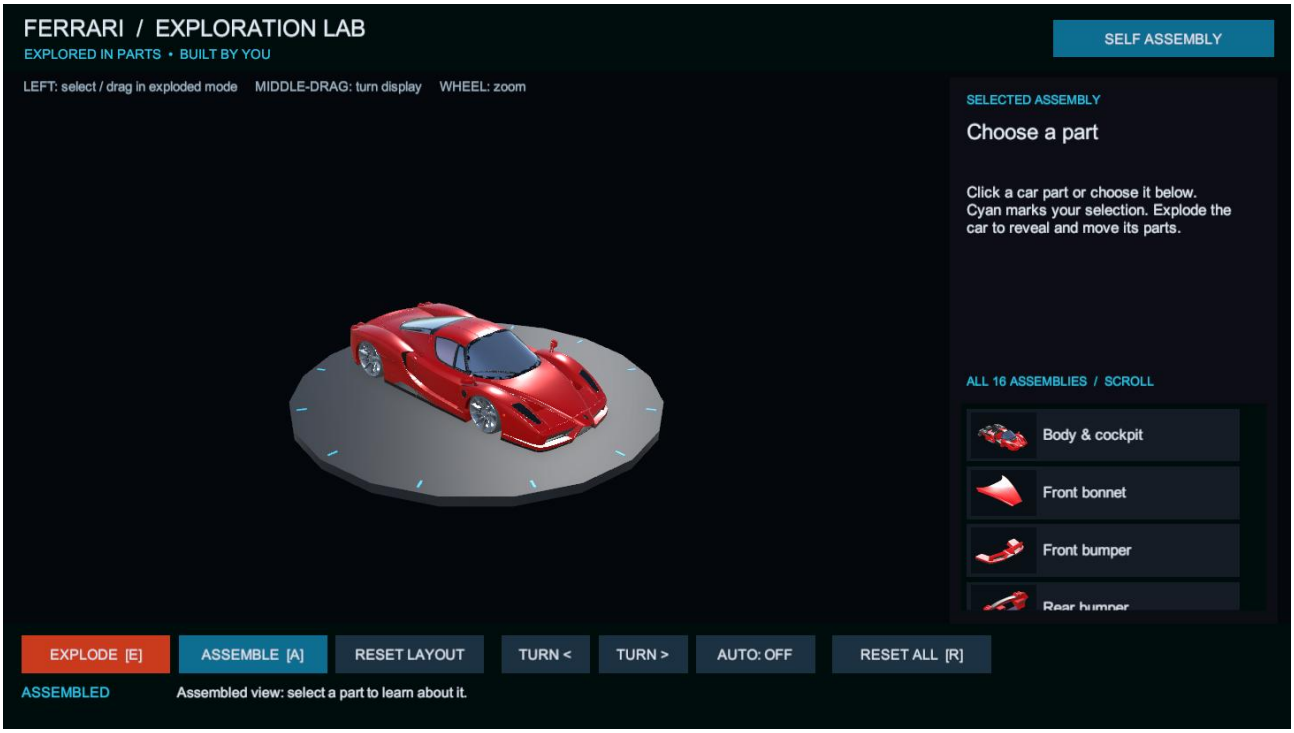
Interactive 3D VR Automotive Educational System

Interactive 3D Exploded View and Guided Assembly of Automotive Components

GIT LINK: https://github.com/PrithviBhende/AR_VR_Project

50 Percent Project Milestone Report

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Ferrari Enzo desktop prototype in Unity. HTC VIVE Focus 3 deployment is planned for the next phase.

This report documents the literature foundation, system design and implemented desktop prototype of our proposed VR learning system. The prototype supports component inspection, exploded views and a guided assembly game. Headset rendering, VIVE controller interaction and educational evaluation remain future work.

1. Introduction

School students often encounter automotive mechanisms through static diagrams. These diagrams show individual parts but offer limited support for understanding how parts occupy space inside a complete vehicle. A reusable interactive model can support component recognition and discussion without requiring access to a physical car or workshop.

Our project presents a Ferrari Enzo model on a display plate in Unity. Learners can rotate and zoom the display, separate major assemblies, select components to read their purposes, and reconstruct the vehicle through guided drag-and-drop placement. The assembled reference poses provide repeatable spatial targets. This is a visual educational model, not a mechanically complete repair simulator.

The intended final display is the HTC VIVE Focus 3 with tracked controllers. We first developed and tested the interaction logic on desktop so that the model structure, educational information and assembly rules could be reviewed before adapting them to VR. The current milestone demonstrates these desktop features; it does not claim a completed headset application.

2. Literature Review

Review scope and selection

The review focuses on automotive education, assembly and maintenance training, instructional design and assessment in VR. Twenty publications were selected through publisher and academic records using terms such as virtual reality automotive training, VR assembly disassembly, immersive learning cognitive load and VR vocational assessment. Priority was given to 2021 to 2026 publications. Two earlier science-learning studies were retained as foundations because they test the assumption that immersion necessarily improves learning [2, 3].

This is a focused project literature review, not an exhaustive systematic review. Publisher abstracts, available article text and DOI metadata support the comparisons. Limitations in the analysis include both study-reported constraints and our assessment of their applicability to school-level automotive learning. Hardware details are included only when confirmed; a particular headset is not assumed from the use of the term VR.

Category	Paper numbers	Count
Foundational or conceptual studies	1 to 3	3
Recent application-oriented studies	4 to 11	8
Technology and methodology studies	12 to 15	4
Application-specific studies	16 to 18	3
Review or survey papers	19 to 20	2
Total	1 to 20	20

Immersion and instructional support

CAMIL describes learning as a relationship between immersion, agency, cognitive and affective processes, and the instructional method [1]. The science-learning comparisons warn that presence and enjoyment should not be treated as evidence of improved understanding [2, 3]. Accordingly, our interface pairs each part with a short purpose and placement clue. A successful assembly is an interaction result; whether it improves component knowledge must be tested separately.

Practice and guidance in assembly

The painting and assembly studies show why interaction should follow learning objectives rather than simply provide free movement [4, 5, 12]. The motor-wiring experiment gives an example of measuring performance on a real task after training, but its small sample does not establish general superiority of VR [6]. Industrial robot and blended-laboratory studies demonstrate repeatable practice and structured exercises [7, 8]. These findings support our restartable assembly mode and motivate a later comparison between guided practice and unaided placement.

Automotive relevance and evaluation

The automotive operator pilot highlights differences between novice and experienced VR users [9]. The vehicle-electrical study combines VR with a teaching strategy, so its outcomes cannot be attributed to the display technology alone [10]. Energy-audit training provides an adjacent automotive-sector example of repeatable procedures, although its content is not vehicle anatomy [11]. The direct automotive laboratory work is closest to our inspection and component-information features [16, 17]. The detailing study links psychological factors to procedural outcomes rather than proving that engagement causes learning [18].

Limits of transfer to our project

Maintenance comparisons show that the best training medium depends on task complexity [13]. Industrial safety research found learning gains without a significant VR advantage over monitor viewing in its first study [14]. Process-based auto-repair assessment suggests tracking errors and decisions, beyond the final completion count [15]. The two reviews identify heterogeneous evidence and unresolved questions about usability, realism and evaluation [19, 20]. We therefore propose a modest, testable educational contribution: part recognition and spatial placement in a guided car model. We do not claim a new assembly algorithm or replacement for workshop training.

3. Literature Review Analysis

The following matrix uses the requested seven columns. Paper numbers link each row to its full reference. Papers 1 to 3 are foundations, 4 to 11 are recent applications, 12 to 15 concern methodology, 16 to 18 are direct automotive studies, and 19 to 20 are reviews.

No.	Authors and year	Research objective	Methodology	VR technology	Key findings	Limitations
1	Makransky and Petersen 2021	Explain how immersion affects learning.	Theoretical synthesis of educational VR research.	Immersive HMD learning; conceptual model.	Presence and agency interact with cognitive load, confidence and learning processes.	Conceptual framework; does not validate this car prototype.
2	Parong and Mayer 2018	Compare science learning in VR and slides.	Media comparison and generative-learning experiments.	Immersive science lesson versus desktop slides.	Instructional support matters; immersive presentation alone does not ensure better knowledge learning.	Science lesson context; not automotive assembly.
3	Makransky et al. 2019	Compare immersive and desktop science labs.	Controlled lab simulation comparison.	HMD and desktop versions of a science simulation.	Greater presence accompanied lower learning in the studied immersive condition.	Task and interface specific; not proof that all VR harms learning.
4	Mulders et al. 2024	Evaluate vocational painting competencies.	Evaluation with 47 apprentices.	Unity painting booth, HTC Vive and adapted spray controller.	Promising usability and support across skill, knowledge and attitude dimensions.	Small sample, self-reports and uncertain transfer to real painting.
5	Wolfartsberger et al. 2023	Assess guidance during assembly training.	64-person comparison of two VR variants and tutor training.	Interactive VR assembly with differing visual guidance.	Guidance changes altered learning and workload; less guidance increased mental demand.	Simplified laboratory task; guidance requires task-specific tuning.

No.	Authors and year	Research objective	Methodology	VR technology	Key findings	Limitations
6	Obermair et al. 2024	Compare electric-motor wiring training.	Two groups of 22 participants total; real wiring test.	VR training versus instruction and printed guidance.	VR-trained participants wired faster; no wiring errors in that group in this experiment.	Small study and experience-dependent results; not whole-car assembly.
7	Pratticò and Lamberti 2021	Investigate robot-operator self-training.	KUKA case study comparing VR and company training.	Immersive robot training with guided tasks.	Demonstrates an industry-linked self-tuition workflow and comparative evaluation.	Robot-specific tasks; physical cable and tool manipulation remain concerns.
8	Antonelli et al. 2023	Support blended robotics laboratories.	JANUS laboratory development and workshop evaluation.	VR robotics laboratory linked to ViLLE learning platform.	Students responded favourably; laboratory exercises can be repeated safely.	Learning performance was measured indirectly; contexts differed across sites.
9	De Vestel and Zogopoulos 2026	Evaluate automotive assembly support.	Pilot study of 11 operators at varied support levels.	VR assembly prototype with controller-based interaction.	Positive experience; novices were slower and support design affected physiological measures.	Small pilot; bodily ownership needed improvement.
10	Liao et al. 2023	Evaluate VR vehicle-electrical teaching.	Eight-week pre/post classroom comparison, 43 and 36 students.	VR simulation games embedded in instructional strategies.	VR-integrated guided instruction supported practical inquiry and learning in the case study.	Single-school setting; VR and teaching strategy effects are intertwined.

No.	Authors and year	Research objective	Methodology	VR technology	Key findings	Limitations
11	Millán et al. 2022	Train automotive SMEs in energy efficiency.	European company case studies within E2DRIVER.	VR switchboard measurement practice within blended training.	Shows repeatable procedural practice with feedback as part of a wider training approach.	Energy auditing rather than car-part anatomy; VR effect not isolated.
12	Mulders 2022	Design coherent vocational VR instruction.	Design-based research using 4C/ID and a ten-step approach.	Vehicle-painting VR tasks and supporting information.	Provides a blueprint linking tasks, information and repeated practice.	Design study; limited time in VR and implementation constraints.
13	Liu et al. 2022	Compare maintenance training media.	60 male trainees across three crane-maintenance task levels.	VR, AR and traditional equipment/video platforms.	Traditional training performed better for simple tasks; AR benefited more complex tasks.	All-male sample and crane context; outcomes depend on task complexity.
14	Mousavi et al. 2023	Explain variation in VR training learning.	Two studies; comparison and structural equation modelling.	Industrial maintenance/safety VR; HTC Vive Pro 2 and monitor video.	Knowledge and confidence rose, but VR did not significantly outperform monitor viewing in Study 1.	Associations do not establish causal superiority; not school automotive instruction.
15	Xiang et al. 2022	Evaluate auto-repair learning processes.	KSA index model and machine-learning process evaluator.	VR theoretical training evaluation auto-repair system.	Demonstrates evaluation using learner process data, not completion alone.	Evaluation model is task-specific; our prototype lacks this analytics layer.

No.	Authors and year	Research objective	Methodology	VR technology	Key findings	Limitations
16	Hernández-Chávez et al. 2021	Teach engine components and operation.	Development of an automotive teaching laboratory.	Oculus Rift S, Leap Motion and four-stroke engine models.	Combines manipulation, assembly/disassembly, descriptions and engine animation.	Engineering-student context and engine scope; not VIVE Focus 3 validation.
17	Makarova et al. 2023	Support automotive-service knowledge transfer.	Readiness survey of 344 students; laboratory use with 109.	Automotive virtual laboratories with practical work scenarios.	Reported improved interest and training performance in the evaluated context.	University context; does not establish transfer to school pupils or this interface.
18	Tai et al. 2022	Study car-detailing procedural learning.	Single-group study with 143 technical students; repeated tests and SEM.	VR car-detailing practice with learning questionnaires.	Flow predicted procedural accuracy and quality; self-efficacy related to interest and anxiety.	No randomized control group; predictive relationships are not causal proof.
19	Di Pasquale et al. 2024	Synthesize VR assembly/disassembly training.	Systematic literature review of operator-training studies.	Multiple VR devices, applications and assessment methods.	Promising but limited evidence; objective and subjective measures both matter.	Heterogeneous tasks and measures limit broad comparisons and generalization.
20	Singhaphandu and Pannakkong 2024	Review industrial virtual-training technology.	Review of industrial VR, AR and e-learning studies.	Varied immersive and non-immersive training systems.	Identifies realism, cost, usability, technical complexity and assessment challenges.	Broad technology mix; not a controlled test of this educational system.

4. Identification of Research Gap

The reviewed work includes engine manipulation, automotive service instruction, vocational practice and assembly training. Our contribution is not the first automotive VR environment. The project gap is a specific instructional and implementation need: an accessible school-level activity that connects component identity, purpose and spatial placement within the same vehicle, with reversible exploration and forgiving guided assembly [5, 9, 16, 19].

Many cited evaluations involve university students, apprentices or industrial operators. Their results do not establish effectiveness for our intended school audience. We must evaluate whether combining exploded inspection with a thumbnail inventory and placement hints improves part identification and spatial recall, compared with static diagrams or desktop inspection alone. This remains a research question, not a demonstrated learning gain.

5. Problem Definition and Project Objectives

Problem definition: students need a clear way to relate major automotive components to their positions and basic functions without needing a physical dismantled vehicle. The system should make parts visible, provide concise explanations and allow repeated reconstruction while avoiding the complexity of a full mechanical CAD simulation.

Provide an assembled and exploded vehicle view with 16 identifiable logical assemblies.

Display each selected component’s image, purpose and placement clue.

Offer a repeatable assembly exercise with 15 movable candidates and a fixed body.

Accept a matching part near its target and animate its position and orientation into place.

Open already-placed access panels when learners place selected interior components.

Adapt selection, movement and UI to HTC VIVE Focus 3 controllers in the next phase.

Evaluate component identification, spatial placement, usability and comfort without assuming educational superiority.

Scope boundary

The available asset does not supply independently verified transmission, battery, braking or suspension assemblies. Those broader synopsis goals and piston/crankshaft working demonstrations are not completed at this milestone. Current animations show separation, snapping and access-panel movement, not engine combustion or drivetrain physics.

6. Requirements

Requirement type	Selected or proposed requirement	Status
VR hardware	HTC VIVE Focus 3 and left/right tracked controllers	Planned
Development platform	Windows computer running the current Unity project	In use
Game engine	Unity 6000.4.11f1 and URP rendering	In use
Programming	C#; Unity Input System and uGUI	In use
3D preparation	Blender 5.1.2; prepared FBX and Unity prefab	In use
VR framework	Unity OpenXR with VIVE OpenXR controller support; XR interaction layer	Planned
Testing	Editor interaction checks now; standalone, headset and learner tests later	Partly complete

Unity retains the current desktop implementation and Blender provides an editable source for grouping, scale correction and future simplification. An FBX import preserves the part hierarchy and makes the same saved poses available to inspection and assembly. The supplied OBJ remains the original graphical source; it is not recreated as an engineering CAD assembly.

HTC VIVE Focus 3 integration requirements

The official Focus 3 specifications list 2448 by 2448 pixels per eye, a 90 Hz refresh rate and an all-in-one design [21]. The official VIVE OpenXR documentation includes Focus 3 controller support for Android all-in-one VR and PC VR with streaming [22, 23]. Standalone Android is the intended deployment direction, but the final route depends on available equipment, SDK compatibility and measured performance.

Before adding XR packages, we will verify the supported Unity, OpenXR, VIVE plugin and headset firmware combination in a separate project checkpoint. The current desktop editor version alone does not prove SDK compatibility. VR selection will use controller rays or grab interactions, with world-space information panels. Desktop camera zoom will not be mapped directly to artificial head-camera movement.

Functional and quality requirements

Stable part IDs must prevent duplicate placement. All modes must restore original local poses on reset. Inventory scrolling must be independent of scene zoom. Dragging and snapping must lock conflicting turntable/camera input. Parts must remain above the base in the authored exploded layout. The headset phase must also verify readable text, comfortable panel placement, controller reach, frame timing and tracking-loss recovery. At 90 Hz, one display interval is approximately 11.1 ms; this is a target budget, not a measured prototype result.

7. System Design

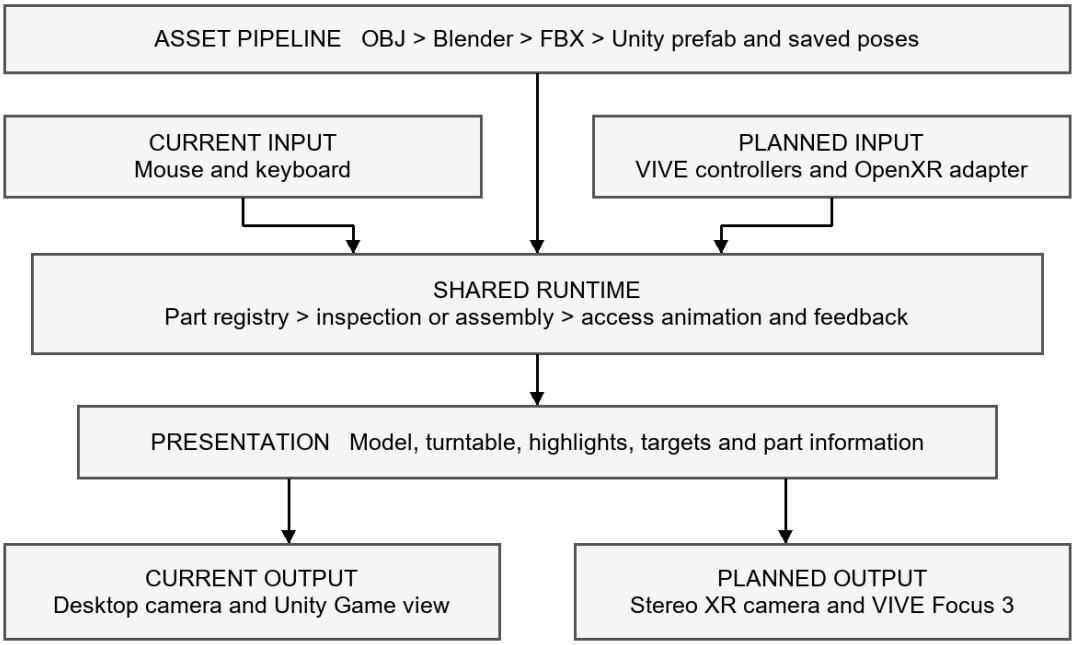


Figure 1. System architecture. The planned input and output branches describe future headset integration.

The common rotating root contains the plate and all assemblies. Saved part positions, rotations and scales are local to that root, so rotating the display does not invalidate assembly targets. Inspection and assembly share the registry and presentation layer, but only the active mode owns selection and drag input.

Component	Responsibility
FerrariPart	Stable ID and original assembled local pose
FerrariInspectionController	Selection, exploded poses, model movement and camera
FerrariAssemblyController	Available/placed state, target checks and snap lifecycle
FerrariLoopingInventory	Recycles visible rows through remaining part IDs
Presentation and access animation	Thumbnails, purposes, clues and panel opening/closing

Use cases and interaction flow

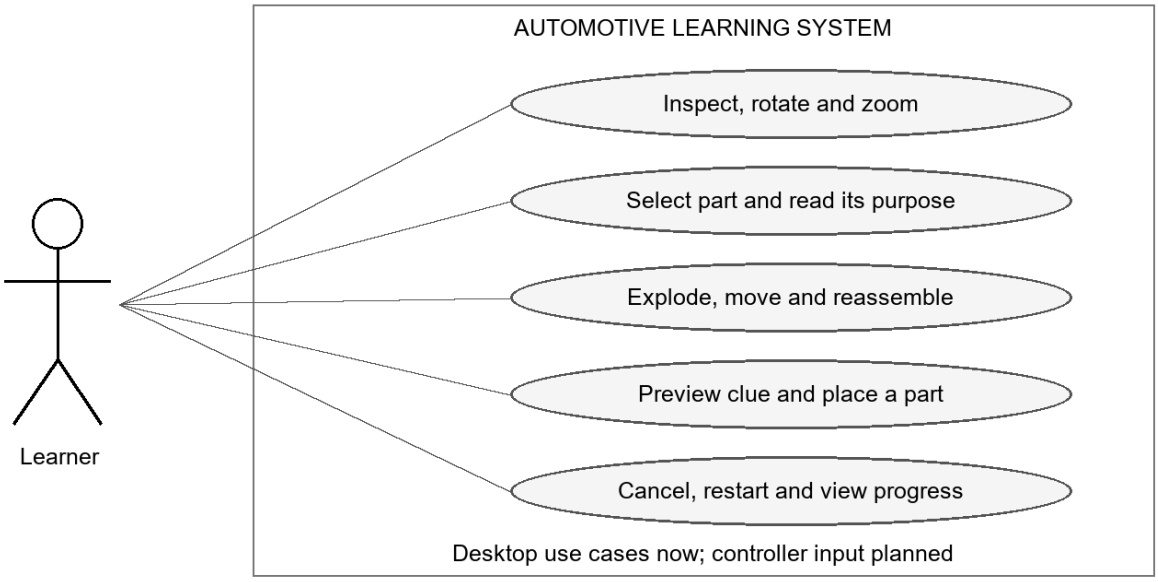


Figure 2. Use-case view. Learner actions are implemented on desktop; controller-based access is planned.

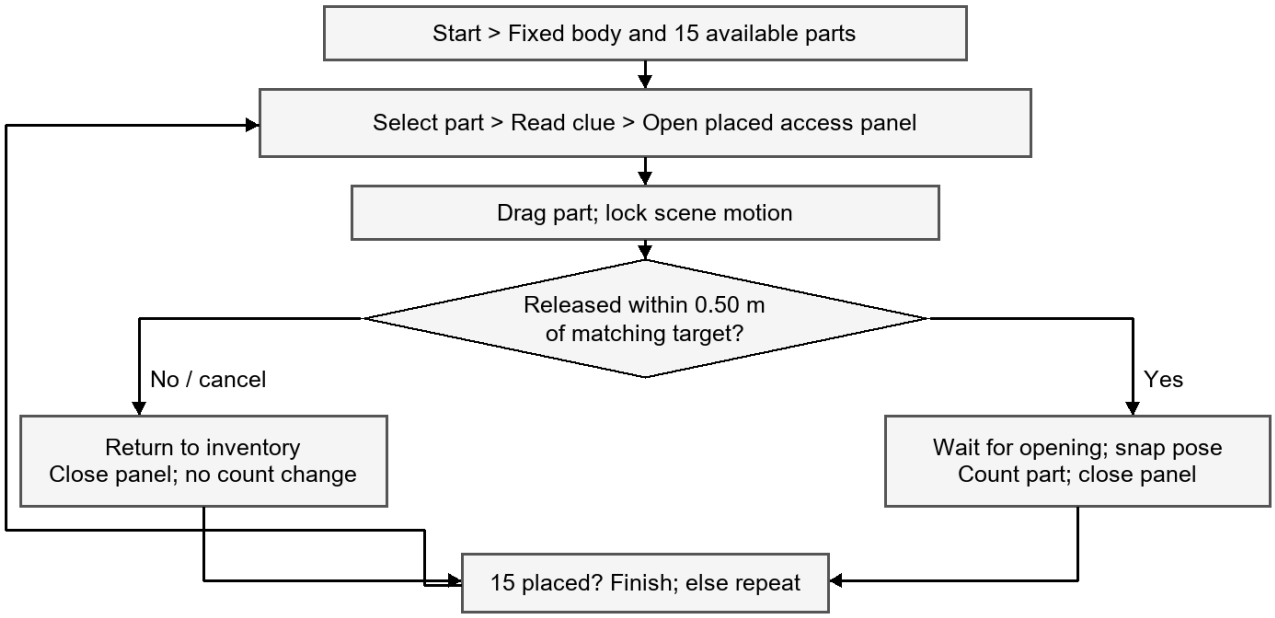


Figure 3. Assembly activity flow. Proximity is checked against the selected part's own anchor, not arbitrary geometry.

8. Initial VR Prototype

The initial prototype is presently a desktop implementation of the proposed VR learning environment. It is demonstrated in Unity Play Mode through FerrariInspection.unity. It provides the important inspection and assembly interactions, while headset display and controller input will be integrated later.

Asset preparation and model structure

The supplied Ferrari Enzo OBJ was inspected and prepared into 16 logical groups. The original source remains unchanged. Paired wheel geometry was separated into four wheel groups, the car was scaled to metres, and the vehicle was positioned above the plate. Replacement materials approximate the appearance because the original material and texture files were not supplied.

Group type	Assemblies
Fixed reference	Body and shared cockpit structure
Exterior and access panels	Front bonnet, front/rear bumpers, left/right doors, rear engine cover, rear wing
Interior and power unit	Left/right seats, steering wheel, engine
Wheels	Front left/right and rear left/right

The Unity prefab has 299 mesh renderers and selection colliders. The prepared source contains 700,735 polygon faces before FBX triangulation. This detail level still requires profiling and likely simplification for standalone VR. The logical groups are educational selections, not verified mechanical subassemblies.

Inspection and guided assembly

Inspection mode animates between assembled and exploded poses. Learners can select a mesh or its right-side list entry, read its information and drag it while exploded. Middle-mouse horizontal movement rotates the plate and all parts about one vertical axis. Wheel input over the scene zooms; wheel input over the list scrolls inventory. The zoom sensitivity was increased in the latest desktop update.

Self assembly fixes the body and offers 15 remaining candidates. Recycled list rows repeat those IDs without creating extra model instances. Selecting an entry previews its image and placement clue. Dragging moves the corresponding part in a target-depth plane. Releasing within 0.50 m of its matching anchor animates the exact saved position, rotation and scale; an invalid release returns it to inventory. Completion occurs at 15 placed parts. Progress survives mode changes only within the current Play session.

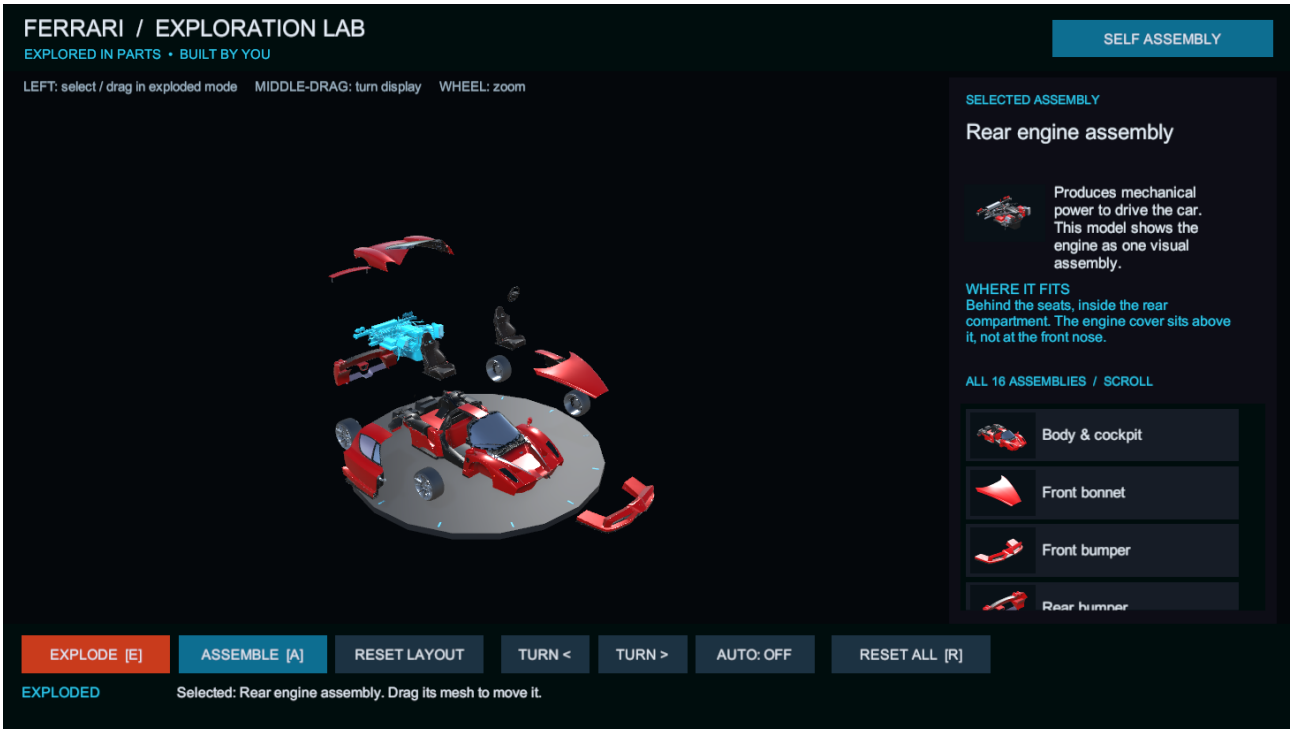


Figure 4. Actual Unity exploded-view checkpoint. The plate and separated groups share the rotating root.

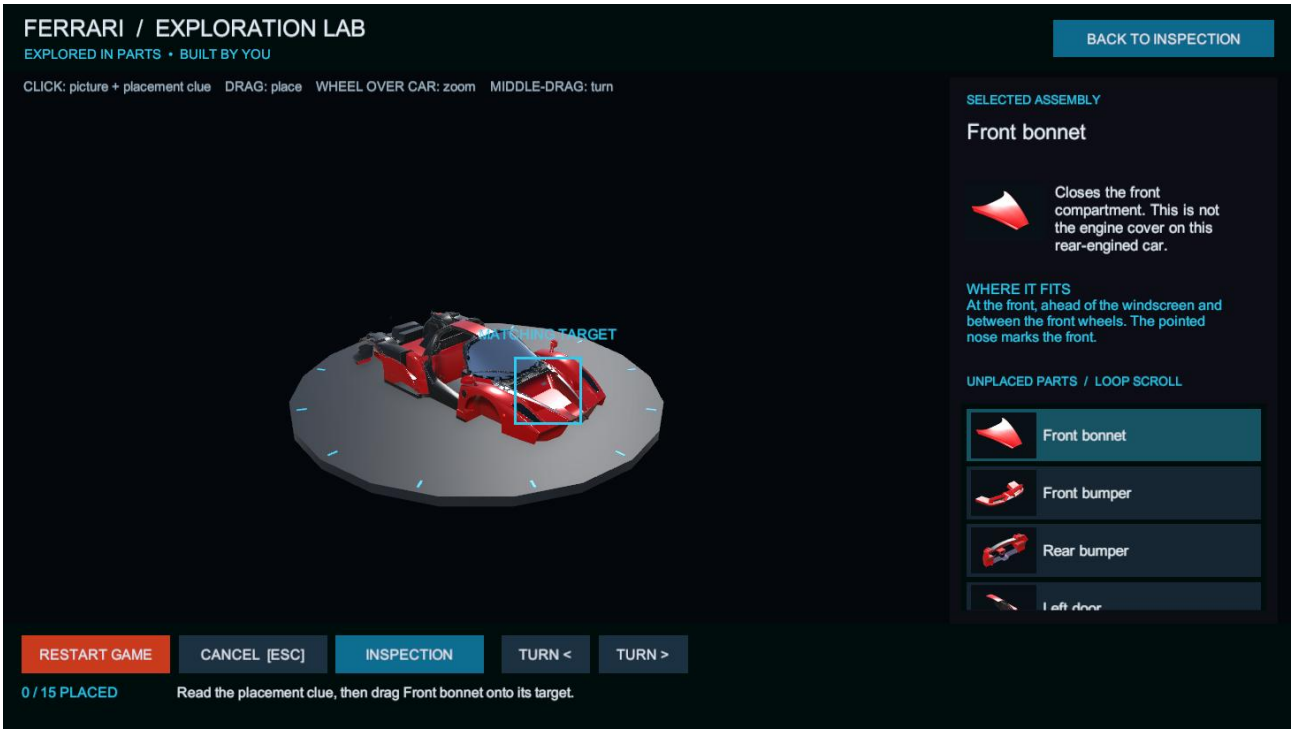


Figure 5. Actual right-side thumbnails and component information. Images depict the prepared model groups.

The interface separates component selection from placement, allowing the learner to read the purpose and location clue before taking the part. Reset functions recover the authored model layout.



Figure 6. Dragging a candidate toward its matching target in the desktop assembly game.

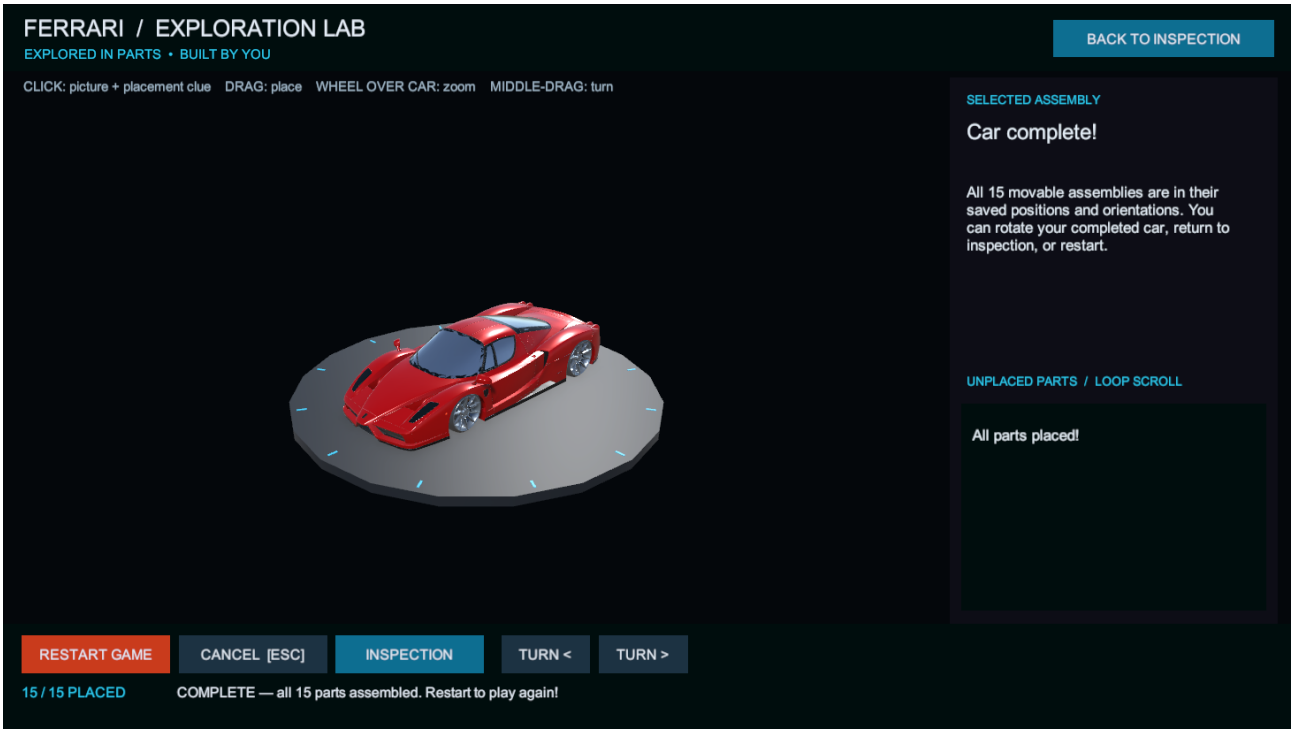


Figure 7. Completed assembly checkpoint showing all 15 movable candidates placed.

Green target feedback indicates proximity acceptance. The snap is deliberately forgiving and supports learning spatial relationships; it does not test fasteners, tolerances, torque, physical interference or repair safety.

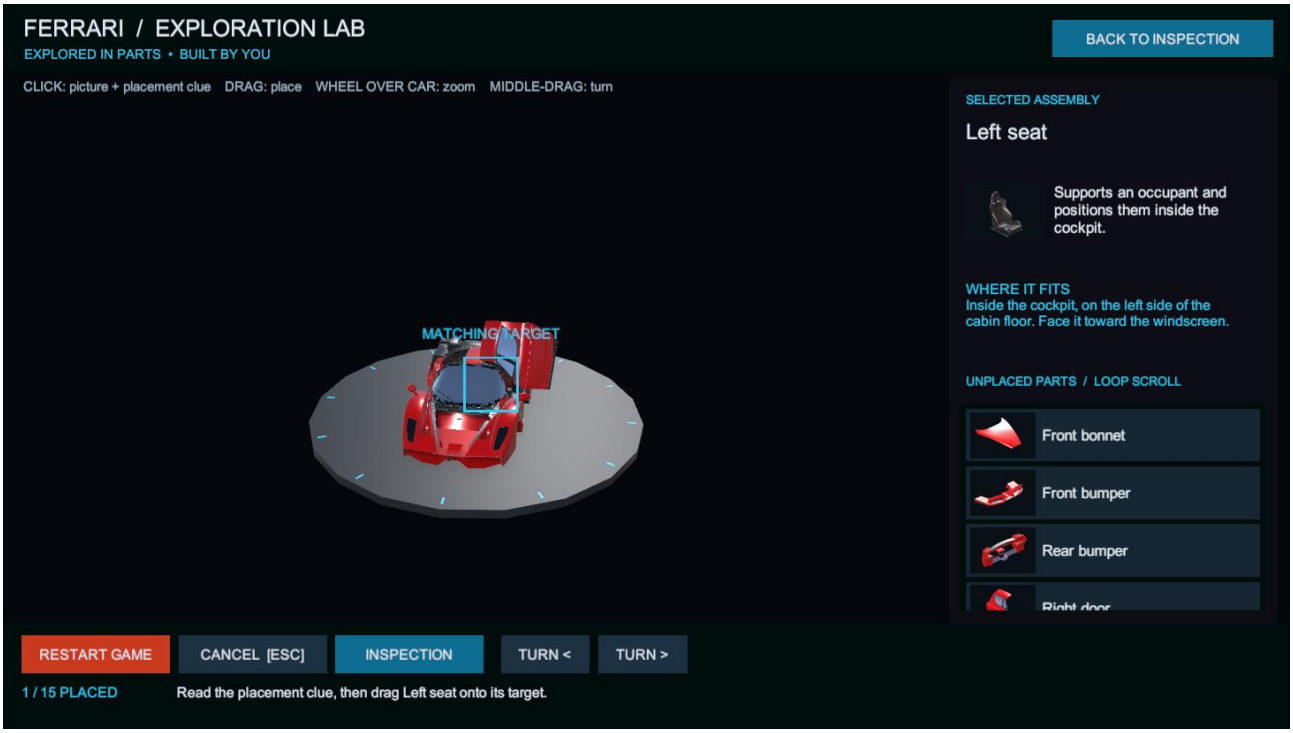


Figure 8. An already-placed left door opens during left-seat access.

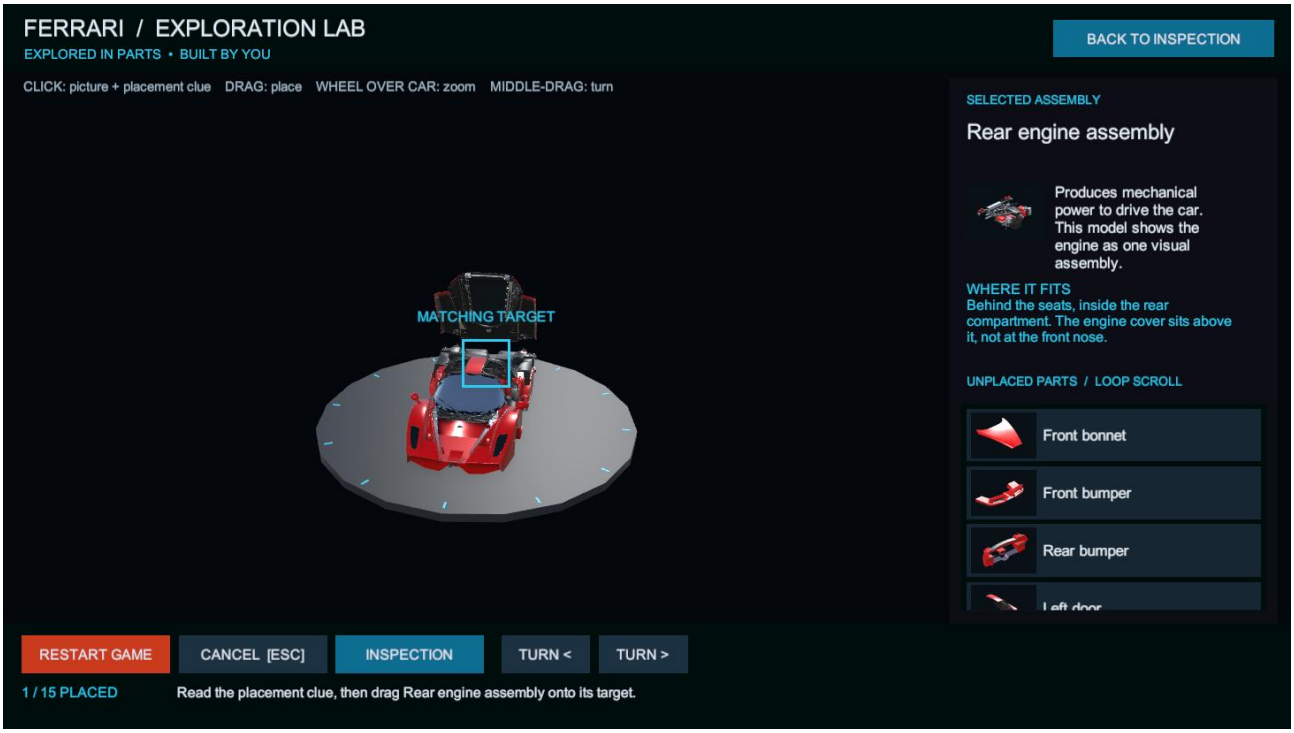


Figure 9. The rear engine cover opens for engine placement. The engine is at the rear of this vehicle.

Placed left/right doors open for their corresponding seats; the left door also opens for the steering wheel. The rear cover opens for the engine, and the front bonnet lifts for front-bumper service access. Panels close after placement, cancellation or a missed drop. Opening uses illustrative hinges inferred from model bounds because the source has no measured hinge rig. These movements are not verified Ferrari mechanical motion.

9. First 50 Percent Project Milestone

The milestone combines the research and design foundation with a working desktop demonstration. The teacher's distribution below is an allocation of expected project work, not a measured percentage of implementation or a claim that all VR requirements are complete.

Project component	Guideline allocation	Milestone evidence
Problem definition and objectives	5%	Defined learner problem, scope and objectives
Literature review of 20 papers	10%	Comparative matrix and source references
Requirements and technology selection	5%	Current tools and planned Focus 3 route
System design and architecture	5%	Architecture, use cases and activity flow
Initial VR prototype	25%	Desktop inspection/assembly implemented; headset phase pending
Total milestone allocation	50%	Submission target; faculty assessment determines acceptance

Recorded functional verification

Editor verification suite	Passing checks	Failures
Inspection and exploded-view regression	38	0
Assembly and thumbnail interface	81	0
Access-panel animation	63	0
Total	182	0

These counts come from recorded Unity Play Mode runs on 9 October 2026. The checks combine UI/mouse events with calls to the interaction methods and cover pose restoration, selection, floor constraints, looping inventory, valid/invalid placement, progress, restart and access-panel recovery. They are automated assertions, not 182 users or an educational experiment. The latest zoom-only sensitivity change followed these runs; the full recorded suite is not represented as a post-change rerun.

Outstanding acceptance work

A standalone player build, headset deployment, XR input, mobile performance profiling and classroom study remain pending. Current dragging is transform-based; parts may visually pass through other geometry. The 50% demonstration should therefore be presented as the functional desktop foundation for the final VR system, with the headset limitation stated during assessment.

10. Documentation

Development decisions and difficulties

Replacing the early placeholder vehicle with the supplied Ferrari improved visual identification, but required grouping the source into useful educational parts. Missing textures required approximate materials. Stable IDs and saved local poses were retained across all checkpoints so that inspection, assembly and later XR input can share the same data contract.

Early interface readability and layout issues led to a right-hand component panel with actual-part thumbnails and short placement hints. A shared parent transform keeps loose parts aligned with turntable rotation. Assembly locks camera and rotation while dragging to avoid shifting the target plane. Access animations address interior visibility but remain illustrative rather than collision-aware service procedures.

Individual contribution record

The project is organized into four sequential work packages: model preparation; inspection and exploded interaction; assembly and access animations; documentation and future VR integration. Prithvi Bhende is recorded as the model-preparation owner in the project handoff. The remaining personal allocations should be confirmed by the team before submission; completed package evidence alone is not proof of an individual student's contribution.

Work package	Completed or planned work	Contribution record
1 Model preparation	Grouped Ferrari, corrected scale, prepared FBX/prefab and asset evidence	Prithvi Bhende; recorded handoff owner
2 Inspection	Explode/assemble, selection, descriptions, rotation and zoom	Completed package; confirm member allocation
3 Assembly	Inventory, target snap, progress, thumbnails and access animations	Completed package; confirm member allocation
4 Documentation and XR	This report; future Focus 3 integration and evaluation	Report prepared; confirm owner; XR pending

Reproducibility and submission evidence

The project scene is My project/Assets/Ferrari/Scenes/FerrariInspection.unity. The editable model is FerrariSource/FerrariEnzo_Prepared.blend. Evidence captures are retained under Evidence, and verification logs under FerrariSource. The Assets, Packages and ProjectSettings folders form the Unity handoff; Library, Temp and other generated caches are not required for a teammate's import. Record the editor/package versions when sharing a checkpoint and run one verification suite at a time.

Confirm the supplied model's license and attribution with its provider before public distribution. The prototype and report do not imply endorsement by Ferrari or HTC. No public repository, video or classroom result is claimed in this report.

11. Expected Outcome

At this milestone, learners can explore a recognizable vehicle, distinguish major component groups and practise placing them within a complete model. The documented desktop checks support functional feasibility. They do not yet demonstrate headset performance, comfort, knowledge retention or transfer to real vehicle work.

Remaining implementation sequence

Preserve the desktop checkpoint, verify SDK compatibility and simplify/profile the asset for the chosen headset deployment route.

Add an XR camera rig, VIVE Focus 3 controller actions, ray selection and comfortable world-space information panels. Keep saved poses and assembly state independent of the input device.

Replace pointer-depth dragging with an appropriate VR grab or ray-placement interaction. Test controller cancellation, tracking loss and access-panel behavior on the actual headset.

Measure frame timing, readable text, model scale and comfort on Focus 3. Adjust rendering resolution and asset complexity based on measurements.

Add selected working-system demonstrations only when the required internal geometry and explanatory content are available.

Run a supervised learner evaluation, analyse results and revise hints and difficulty before the final submission.

Proposed educational evaluation

Use short pre- and post-session questions covering component names, basic functions and spatial relationships. Record assembly completion time and wrong-drop counts separately from questionnaire scores. Compare a static-diagram or desktop condition where feasible, and include a delayed recall measure if time permits. Ask learners about readability, ease of interaction and discomfort. These are planned measures; no participant results have been collected or claimed.

For school pupils, obtain institutional approval and appropriate guardian consent, follow headset age and safety guidance, provide adult supervision and stop immediately if discomfort occurs. Store only necessary anonymized learning data. Educational conclusions must reflect the sample, comparison condition and observed results rather than the novelty of VR.

Conclusion

The project has a working desktop foundation for exploded inspection and guided assembly, supported by a focused literature review and a defined architecture. HTC VIVE Focus 3 integration is the next technical stage. The final system should be judged on accurate explanations, usable interaction and measured learning outcomes, while remaining a supplement to classroom instruction rather than a substitute for mechanical workshop training.

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References use APA-style entries. Numbers provide cross-references to the literature matrix and report text.

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Appendix A Selected Unity Code Snippets

These C# excerpts document the desktop prototype described in Sections 7 and 8. Formatting and line breaks have been adjusted for readability; the executable tokens are checked against the project scripts. The excerpts use fields and helpers from their containing classes and are not standalone programs. No headset integration code is claimed.

A1 Saved poses and exploded view

Each component stores its assembled pose in its parent coordinate system. Restoring this pose remains valid when the common turntable rotates.

Source: Assets/Ferrari/Scripts/FerrariPart.cs; starts at line 15.

```
public void RestoreAssembledPose()
{
    transform.localPosition = assembledLocalPosition;
    transform.localRotation = assembledLocalRotation;
    transform.localScale = assembledLocalScale;
}
```

The inspection transition interpolates each component toward its exploded location or its original assembled position. The enclosing Update method supplies a smoothly increasing transition value.

Source: Assets/Ferrari/Scripts/FerrariInspectionController.cs; starts at line 128.

```
void ApplyPose(float t)
{
    for (int i = 0; i < parts.Length; i++)
    {
        var p = parts[i];
        p.transform.localPosition = Vector3.Lerp(
            starts[i],
            IsExploded ? exploded[p] : p.assembledLocalPosition, t);
        p.transform.localRotation = Quaternion.Slerp(
            rotations[i], p.assembledLocalRotation, t);
        p.transform.localScale = p.assembledLocalScale;
    }
}
```

Appendix A2 Dragging and Matching Targets

The candidate is moved on a plane through its own target depth. Parent-space conversion keeps movement consistent with display rotation. The acceptance check uses the selected part's saved anchor and the 0.50-metre radius, rather than whichever surface the pointer happens to touch.

Source: Assets/Ferrari/Scripts/FerrariAssemblyController.cs; starts at line 47.

```
public Vector3 TargetPosition(FerrariPart p) =>
   p.transform.parent.TransformPoint(p.assembledLocalPosition);
```

Source: Assets/Ferrari/Scripts/FerrariAssemblyController.cs; starts at line 28.

```
public bool CanSnap =>
   CurrentPart && CurrentPart.gameObject.activeSelf &&
   Vector3.Distance(CurrentPart.transform.position,
      TargetPosition(CurrentPart)) <= SnapRadius;
```

The following block is from DragPointer. Bounds correction keeps the held mesh above the plate; it does not provide collision-constrained mechanical assembly.

Source: Assets/Ferrari/Scripts/FerrariAssemblyController.cs; starts at line 93.

```
var ray = inspection.viewCamera.ScreenPointToRay(screenPosition);
if (!dragPlane.Raycast(ray, out float enter)) return;
var local = inspection.displayRoot.InverseTransformPoint(
   ray.GetPoint(enter));
local.x = Mathf.Clamp(local.x, -5, 5);
local.z = Mathf.Clamp(local.z, -5, 5);
local.y = Mathf.Clamp(local.y, .16f, 4.5f);
CurrentPart.transform.position =
   inspection.displayRoot.TransformPoint(local);
var bounds = inspection.BoundsOf(CurrentPart);
if (bounds.min.y < .16f)
   CurrentPart.transform.position +=
      Vector3.up * (.16f - bounds.min.y);
```

Release validates that the pointer is in the car viewport and CanSnap is true. Otherwise CancelDrag returns the part to the inventory without increasing the placed count.

Appendix A3 Animated Snap and Progress

After an accepted release, the coroutine waits for any required access panel to open. It then interpolates local position and orientation over approximately 0.4 seconds. Exact pose restoration avoids accumulated animation error; adding the stable ID records successful placement.

Source: Assets/Ferrari/Scripts/FerrariAssemblyController.cs; starts at line 113.

```
IEnumerator SnapIntoPlace()
{
   var p = CurrentPart;
   var from = p.transform.localPosition;
   var rotation = p.transform.localRotation;
   while (!Access.ReadyForPlacement) yield return null;
   Status = "Placing " + p.displayName + "...";
   for (float t = 0; t < 1; t += Time.deltaTime / .4f)
   {
      float ease = Mathf.SmoothStep(0, 1, t);
      p.transform.localPosition = Vector3.Lerp(
         from, p.assembledLocalPosition, ease);
      p.transform.localRotation = Quaternion.Slerp(
         rotation, p.assembledLocalRotation, ease);
      yield return null;
   }
   p.RestoreAssembledPose();
   placed.Add(p.partId);
   CurrentPart = null;
   PreviewPart = null;
   IsSnapping = false;
   snap = null;
   Access.Close();
   Status = IsComplete
      ? "COMPLETE — all 15 parts assembled. Restart to play again!"
      : "Placed " + p.displayName + ". Choose the next part.";
   RefreshInventory();
}
```

The placed collection is a HashSet of part IDs. The inventory is refreshed from candidates that are not placed and are not currently held, so repeated display rows do not create duplicate physical components.

Appendix A4 Looping Inventory and Drag Events

The inventory reuses a fixed pool of visible rows. Modulo arithmetic wraps the starting index through the remaining candidates in either scrolling direction. Row event handlers forward the same selected component to the assembly controller.

Source: Assets/Ferrari/Scripts/FerrariLoopingInventory.cs; starts at line 33.

```
public void ScrollBy(int steps)
{
   if (assembly.IsBusy || assembly.Available.Count == 0) return;
   int count = assembly.Available.Count;
   StartIndex = ((StartIndex + steps) % count + count) % count;
   RefreshRows();
}
```

RefreshRows assigns each visible row to a remaining ID and loads that component’s thumbnail. The following loop statement is the part-index mapping excerpt.

Source: Assets/Ferrari/Scripts/FerrariLoopingInventory.cs; starts at line 45.

```
var p = count == 0 ? null
   : assembly.Available[(StartIndex + i) % count];
rows[i].part = p;
```

Source: Assets/Ferrari/Scripts/FerrariInventoryItem.cs; starts at line 15.

```
public void OnBeginDrag(PointerEventData e)
{
   if (e.button == PointerEventData.InputButton.Left)
      ownsDrag = inventory.assembly.BeginDrag(part, e.position);
}
```

Source: Assets/Ferrari/Scripts/FerrariInventoryItem.cs; starts at line 17.

```
public void OnEndDrag(PointerEventData e)
{
   if (ownsDrag) inventory.assembly.Release(e.position);
   ownsDrag = false;
}
```

Appendix A5 Turntable Rotation and Scroll Zoom

Rotation is applied to the common display root about its local vertical axis. The same root contains the plate and every car component. Input is blocked while placement owns a drag or snap, so the target plane cannot shift under a held part.

Source: Assets/Ferrari/Scripts/FerrariInspectionController.cs; starts at line 138.

```
public void Turn(float degrees)
{
   if (!IsDragging && !pendingDrag &&
      !(Assembly && Assembly.IsBusy))
      displayRoot.Rotate(Vector3.up, degrees, Space.Self);
}
```

This Update block converts horizontal middle-mouse movement to turntable rotation. OverUI excludes input over the interface.

Source: Assets/Ferrari/Scripts/FerrariInspectionController.cs; starts at line 156.

```
if (mouse.middleButton.wasPressedThisFrame && inView &&
   !OverUI && !pendingDrag && !IsDragging) rotating = true;
if (!mouse.middleButton.isPressed) rotating = false;
if (rotating) Turn(-mouse.delta.ReadValue().x * .3f);
```

Scene scrolling applies proportional zoom with the latest sensitivity value of 0.0045. The clamp limits the zoom range; scrolling over the list is handled separately by inventory events.

Source: Assets/Ferrari/Scripts/FerrariInspectionController.cs; starts at line 181.

```
if (inView && !OverUI && !IsDragging && !pendingDrag)
   zoom = Mathf.Clamp(
      zoom * Mathf.Exp(-mouse.scroll.ReadValue().y * .0045f),
      .35f, 2.2f);
```

The camera update smooths distance changes rather than jumping directly to the newly requested distance.

Source: Assets/Ferrari/Scripts/FerrariInspectionController.cs; starts at line 221.

```
distance = Mathf.Lerp(distance, requestedDistance * zoom,
   1 - Mathf.Exp(-10 * Time.deltaTime));
```

Appendix A6 Opening Placed Access Panels

The request map opens only panels that are already placed. Seats and steering trigger their corresponding doors; engine placement triggers the rear cover. The front bonnet supports an illustrative front-bumper access activity. These pairings are visual educational behavior, not a validated repair procedure.

Source: Assets/Ferrari/Scripts/FerrariAccessAnimator.cs; starts at line 55.

```
public void RequestFor(FerrariPart interior)
{
   string id = interior ? interior.partId : "";
   foreach (var p in panels)
   {
      bool required =
         (p.part.partId == "door_L" &&
            (id == "seat_L" || id == "steering_wheel"))
         || (p.part.partId == "door_R" && id == "seat_R")
         || (p.part.partId == "engine_cover" && id == "engine")
         || (p.part.partId == "bonnet" && id == "bumper_F");
      p.target = assembly.IsActive && required &&
         assembly.IsPlaced(p.part) ? 1 : 0;
   }
}
```

The LateUpdate motion excerpt advances the opening amount using Duration = 0.70 seconds and applies rotation around the inferred hinge in parent space. The surrounding method excludes unplaced panels and restores the exact closed pose when the amount reaches zero.

Source: Assets/Ferrari/Scripts/FerrariAccessAnimator.cs; starts at line 79.

```
p.amount = Mathf.MoveTowards(
   p.amount, p.target, Time.deltaTime / Duration);
if (p.amount <= 0)
{
   p.part.RestoreAssembledPose();
   continue;
}
var rotation = Quaternion.AngleAxis(
   p.angle * Mathf.SmoothStep(0, 1, p.amount), p.axis);
p.part.transform.localPosition = p.pivot +
   rotation * (p.part.assembledLocalPosition - p.pivot);
p.part.transform.localRotation =
   rotation * p.part.assembledLocalRotation;
p.part.transform.localScale = p.part.assembledLocalScale;
```

Close changes panel targets back to zero after placement or cancellation. Restart, mode exit and focus loss additionally restore closed poses immediately. The source OBJ has no measured hinge rig, and the prototype does not simulate panel collision or joint forces.